Suggested problems: 1-21 odd, 25-31 odd

Problem 1: Find all critical points of $f(x, y) = x^2 + y^2$. Classify the points as maximums, minimums, or neither.

The equation

$$\nabla f(x, y) = \langle 2x, 2y \rangle = \langle 0, 0 \rangle$$

implies $x = y = 0$. Thus $(0, 0)$ is a critical point of f. The point $(0, 0, 0)$ is a local minimum, which can be verified by checking the graph.

Problem 2: Find all critical points of $f(x, y) = -\sqrt{x^2 + y^2}$. Classify the points as maximums, minimums, or neither.

The gradient

$$\nabla f(x, y) = \left\langle -\frac{x}{\sqrt{x^2 + y^2}}, -\frac{y}{\sqrt{x^2 + y^2}} \right\rangle$$

is undefined at $(0, 0)$, so this point is a critical point. There is a local maximum at $(0, 0, 0)$, which can be verified by checking the graph.

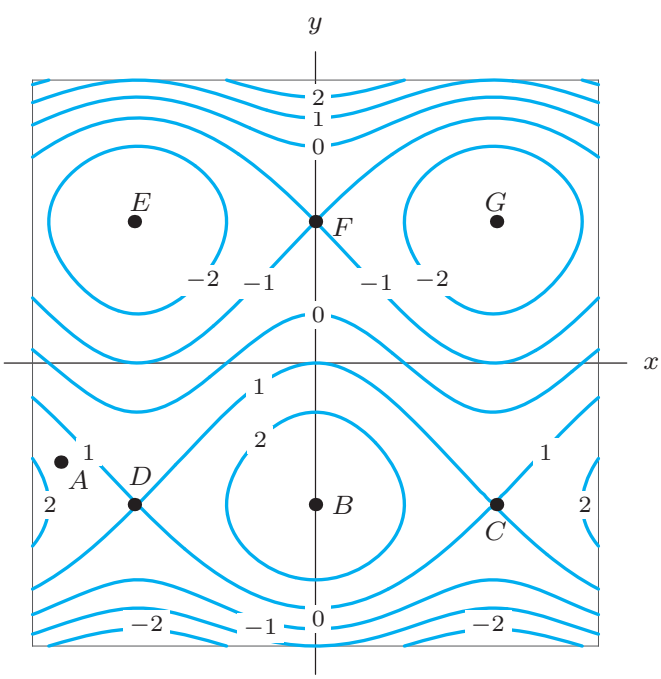
Problem 3: Find all critical points of $f(x, y) = x^2 - y^2$. Classify the points as maximums, minimums, or neither.

The equation

$$\nabla f(x, y) = \langle 2x, -2y \rangle = \langle 0, 0 \rangle$$

implies $x = y = 0$, so $(0, 0)$ is a critical point. There is neither a local maximum nor local minimum at $(0, 0, 0)$, which means it's a saddle point.

Problem 4: Decide whether the points A-G in the contour diagram are critical points. For those that are, classify them as local maximums, minimums, or neither.



A is not a critical point, B is a critical point and a local maximum, C is a critical point and a saddle point, D is a critical point and a saddle point, E is a critical point and a local minimum, F is a critical point and a saddle point, G is a critical point and a local minimum

Problem 5: Find all critical points of $f(x, y) = 5 + 6x - x^2 + xy - y^2$. Classify the points as maximums, minimums, or neither.

We first find critical points:

$$\nabla f(x, y) = \langle 6 - 2x + y, x - 2y \rangle = \langle 0, 0 \rangle$$

implies $x = 4$ and $y = 2$. Then

$$D = \begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix} = 4 - 1 = 3 > 0.$$

Since $f_{xx}(4, 2) = -2 < 0$, we see $(4, 2, 17)$ is a local maximum.

Problem 6: Find all critical points of $f(x, y) = x^2y + 2y^2 - 2xy + 6$. Classify the points as maximums, minimums, or neither.

We first find critical points:

$$\nabla f(x, y) = \langle 2xy - 2y, x^2 + 4y - 2x \rangle = \langle 0, 0 \rangle$$

implies either $x = y = 0$, $x = 2$ and $y = 0$, or $x = 1$ and $y = 1/4$. We have

$$D = \begin{vmatrix} 2y & 2x - 2 \\ 2x - 2 & 4 \end{vmatrix} = 8y - (2x - 2)^2$$

Plugging in $(0, 0)$ gives $D = -4 < 0$, so $(0, 0, 6)$ is a saddle point. Plugging in $(2, 0)$ gives $D = -4 < 0$ again, so $(2, 0, 6)$ is also a saddle point. Plugging in $(1, 1/4)$ gives $D = 2 > 0$. Since also $f_{xx}(1, 1/4) = 1/2 > 0$, this means there is a local minimum at $(1, 1/4, 47/8)$.